Requirements Document

Livelihood Installation Application

(Reuse of E4H platform for Livelihood program)

Document Type	High-Level Requirements (Delta from E4H)
Program	Livelihood
Base Platform	E4H (Energy for Health) – Web + InstallationQC App
Audience	Internal stakeholders familiar with E4H
Version	v0.1 (Draft)

1. Background

The E4H (Energy for Health) platform is currently used to manage installation of solar panels at health facilities. The platform consists of a Web portal — used for master-data ingestion, project setup, field-plan creation, and role assignment — and a mobile application (InstallationQC module) used by field teams to capture installation data.

The organization now intends to use the same platform to manage installations under the Livelihood program. Under Livelihood, a range of machines (solar panels, roti rolling machines, chilli pounding machines, milking machine and similar assets — some solar-powered, some not) are installed at the premises of end users, typically farmers or individual producers, to support income generation.

This document captures the high-level requirements for extending the existing E4H platform to support Livelihood installations, focusing on the changes required from the current E4H implementation.

2. Objective

To extend the existing E4H Web platform and Installation QC App so that the Livelihood program can be managed end-to-end on the same system, while accommodating the program-specific entity model (end user, item code, vendor), the multi-state project structure, the vendor-led field-plan workflow, and the new acknowledgment, handover, and notification requirements.

3. Scope

3.1 In Scope

- Project creation under Livelihood, driven by a Justification Code, and extended to allow a single project to span multiple states.

- Mapping of end users to a project using the Justification Code.
- Assignment of one or more Item Codes (machines / assets) to each end user; each item code carries a flag indicating whether the solution includes solar.
- Vendor master and mapping of vendors to item codes.
- Field-plan creation at end-user × vendor level — one field plan per vendor per end user.
- Role assignment within each field plan: Field Staff (vendor), Field Supervisor (vendor), Installation Reviewer (SELCO).
- Installation QC App workflow with item-code-driven data capture forms (detailed for solar, lightweight for non-solar machines).
- Template-based generation of an Acknowledgment Letter at end-user level, validated via OTP.
- Template-based generation of a Handover Letter using a separate customisable template.
- Automated email notifications to procurement team on role assignment and on completion of installation and handover reports.

3.2 Out of Scope

- **AMC (Annual Maintenance Contract) module:** Available in E4H but not part of this phase. To be discussed and taken up in a subsequent phase (see Section 9).

4. Key Changes from E4H

The Livelihood program introduces a different entity model and a different planning unit. The changes are summarized below.

Aspect	E4H (Current)	Livelihood (New)
Project geography	A project is restricted to a single state.	A project can span multiple states under a single Justification Code.
Recipient of installation	Health facility.	End user (farmer / individual producer), mapped to the project via Justification Code.
Asset per recipient	One solar panel system per facility.	One or more item codes (machines) per end user; the item code itself indicates whether the solution includes solar.
Vendor concept	Not central to the flow.	Each item code is mapped to one vendor. Field plans are organised around vendors.

Field plan unit	Per facility.	Per end user per vendor. An end user with item codes from two vendors will have two field plans.
Roles in a field plan	Field Staff, Field Supervisor, Installation Reviewer (all internal).	Field Staff and Field Supervisor from the vendor; Installation Reviewer from SELCO (Program POC / Projects team).
Acknowledgment & Handover	Not part of the current flow.	OTP-verified Acknowledgment Letter at end-user level; separate Handover Letter — both template-based.
Notifications	Limited / system-internal.	Email on role assignment within a field plan, and email to procurement team on completion of installation and handover reports.

5. Entity Model

The entities and their relationships are as follows:

- **Justification Code → Project:** One Justification Code drives the creation of one project. The project carries the geography (one or more states, with their districts and blocks).
- **Project → End Users:** One project can have many end users. End users are mapped to the project via the Justification Code.
- **End User → Item Codes:** One end user can have one or more item codes. Each item code represents a unique solution (machine / asset) and might have solar integrated or separately added. Currently logic for item code is being developed.
- **Item Code → Vendor:** Each item code is supplied by one vendor.
- **End User × Vendor → Field Plan:** One field plan is created for each unique combination of end user and vendor. If an end user has item codes from two vendors, two separate field plans are created — one per vendor.

Worked Example

1 Justification Code → 1 Project covering 1 or more states → 3 end users in the project. Each end user has 2 item codes; each item code is from a different vendor. Result: each end user has 2 field plans (one per vendor), so the project has 6 field plans in total.

6. High-Level Functional Requirements

6.1 Project Creation (Web)

Performed by the Program POC.

- Enter the Justification Code, and select State(s), District(s), and Block(s). Multiple states are allowed under a single project (**this is a change from E4H, which restricts a project to one state).**
- On saving, the system maps all end users linked to the Justification Code to the project.
- The Program POC should be able to view the list of mapped end users from the project page.

6.2 End User & Item Code Mapping (Web) (Backend)

- Each end user mapped to the project will have one or more item codes (machines / assets) attached to them.
- Each item code is unique and carries an indicator showing whether the solution includes solar.
- Each item code is linked to one vendor.

6.3 Field Plan Creation (Web)

Performed by the Program POC after the project is created.

- Field plans are generated at end-user × vendor level — one field plan per end user per vendor.
- The vendor is assigned to the field plan at the time of creation.
- All item codes of that end user supplied by the same vendor are grouped under the same field plan.
- Within each field plan, three roles are assigned: Field Staff (vendor), Field Supervisor (vendor), and Installation Reviewer (SELCO Program POC / Projects team).
- An email notification is sent automatically to each assigned user when the field plan is created or roles are added / changed.

6.4 Installation QC App

- The vendor's Field Staff sees the field plans assigned to them, with the list of end user names, item codes (machines) to be installed for each end user.
- The installation form rendered in the app is driven by the item code: if the solar flag is set, the detailed solar form is used (consistent with E4H); for non-solar machines, a lightweight form is used capturing serial number, warranty number, and basic installation evidence. This would be essential to track warranty.
- On submission by the Field Staff, the installation moves through the Field Supervisor and then to the SELCO Installation Reviewer for final approval.

6.5 Acknowledgment Letter (End-User Level, OTP-Verified)

- Once the installation(s) under a field plan are completed and approved, the system generates an Acknowledgment Letter for the end user using a pre-configured template.
- Before the letter is finalised, the end user verifies the installation through an OTP — the OTP is sent to the end user and validated in the app / system.
- On successful OTP validation, the Acknowledgment Letter is generated and saved against the field plan / end user.

6.6 Handover Letter

- In addition to the Acknowledgment Letter, the system generates a separate Handover Letter using its own customisable template.
- The Handover Letter is generated post installation, against the end user / field plan. This would also include basic machine operation training that would be provided to the end user by the vendor.
- The Acknowledgment and Handover templates should be configurable on the web platform by an authorised admin.

6.7 Email Notifications

- **Role assignment notification:** When any of the three roles is assigned within a field plan, an automatic email is sent to that user.
- **Completion notification to Procurement:** Once all installation reports and handover reports under a field plan (or as defined during design) are submitted, an automatic email is sent to the procurement team notifying them of completion.

7. Reuse from E4H

To minimize build effort and timelines, the following are expected to be reused with configuration / minor extension only:

- Geography master-data ingestion (State / District / Block).<- can also bring API
- Project creation framework (extended for multi-state and Justification Code).
- Role definition and assignment framework (extended for vendor-side users).
- Installation QC App shell, authentication, and submission workflow.
- Solar-panel installation form (used as-is for item codes with the solar flag).

8. Assumptions

- Livelihood will run on the same instance and codebase as E4H, with program-level segregation.

- End users and Justification Codes are managed as master data and are available in the system before project creation.
- Vendors are managed as master data and are mapped to item codes outside the project creation flow.
- Acknowledgment and Handover letter templates will be configured once and reused; admin users will be able to update the templates as needed.
- Email addresses for assigned roles and for the procurement team are maintained in the system.
- No change is required to the existing E4H flow as a result of these enhancements.

9. Open Points and Future Phases

Future Phase (To Be Discussed Separately)

- **ERP integration** to use the actual justification code, item code, system type mapping rather than manual ingestion.
- **AMC (Annual Maintenance Contract) module:** The AMC module exists in E4H and supports post-installation maintenance tracking. It is not being replicated in this phase. A separate discussion is planned to decide the scope, applicability, and timeline of AMC for Livelihood, given that the asset base (multiple machine types per end user, multiple vendors) differs materially from E4H.